

Polymer Simulation Knowledge Base

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Chapter 1

Polymer Simulation

A practical knowledge base for molecular simulation of polymers. Find curated literature, force-field guidance, simulation software, and polymer modeling workflows.

[PDF build workflow](#)

From atomistic models to coarse-grained melts: a growing, source-linked reference for reproducible polymer simulation.

1.1 Start here

I want to...	Go to
Find foundational papers and a suggested reading path	Literature
Compare simulation packages and parameterization tools	Tools
Read the complete guide as a book	Download the PDF

1.2 What's covered

- **Models:** atomistic, united-atom, and coarse-grained polymer systems
- **Methods:** force-field selection, parameterization, equilibration, and multiscale modeling
- **Applications:** melts, glassy polymers, electrolytes, and nanocomposites
- **Tools:** LAMMPS, GROMACS, RadonPy, and related packages
- **Analysis:** chain statistics, dynamics, entanglement, and material properties

Literature-reported MD results can differ from uniformly generated computational datasets because outcomes depend on the force field, model representation, polymer chemistry, and simulation protocol.

1.3 Repository map

Directory	Contents
papers/	Reading lists, literature notes, and citation collections
tools/	Software, utilities, installation, and configuration notes

1.4 Build the PDF

The PDF combines the Markdown guides into one document. Install [Pandoc](#) and XeLaTeX, then run:

```
make
```

See `make help` for available targets. The PDF is also built automatically on pushes and pull requests to `main`.

1.5 Contributing

Add material to `papers/` or `tools/` where it fits. Keep notes concise, link claims to their sources, and include dependencies and example usage when adding code. When adding a new Markdown chapter, add it to `SOURCES` in the `Makefile` so it appears in the PDF.

Chapter 2

Papers

Use this folder for literature notes, reading lists, citation collections, and short paper summaries related to polymer simulation.

2.1 VIP Paper

1. High-Throughput Molecular Dynamics Simulations and Validation of Thermophysical Properties of Polymers for Various Applications, ACS Appl. Polym. Mater. (2021) 3 (2): 620–630., <https://doi.org/10.1021/acsapm.0c00524>
2. Paudel, A., Shakya, A., Nag, S. et al. A structured dataset of polymers in molecular dynamics simulations extracted by LLM-assisted text mining. Sci Data (2026). <https://doi.org/10.1038/s41597-026-08045-x>
3. Macromolecules (2019) 52 (3): 755–786. <https://doi.org/10.1021/acs.macromol.8b01836>, Modeling and Simulations of Polymers: A Roadmap
4. Chew, A.K., Afzal, M.A.F., Kaplan, Z. et al. Leveraging high-throughput molecular simulations and machine learning for the design of chemical mixtures. npj Comput Mater 11, 72 (2025). <https://doi.org/10.1038/s41524-025-01552-2>
5. Atomistic polymer modeling: Recent advances and challenges in building and parametrization workflows HN Turney, M Matta - Macromolecules, 2025

2.2 Benchmark

1. Benchmarking Classical Molecular Dynamics Simulations for Computational Screening of Lithium Polymer Electrolytes, Macromolecules (2025) 58 (13): 6732–6742. <https://doi.org/10.1021/acs.macromol.5c00930>

2.3 High-Citation and Classic Reading List

Citation counts are approximate screening signals, because Google Scholar, ACS, Scopus, Crossref, OUCI, and other databases use different coverage. Counts below were checked from publicly visible pages on 2026-05-20.

Year	Paper	Why it matters	Citation signal
1985	Theodorou & Suter, <i>Detailed molecular structure of a vinyl polymer glass</i>	Early atomistic modeling of amorphous polymer glass structure; useful for packing, free volume, and glassy polymer model-building ideas.	ACS: cited by ~823 publications.
1990	Kremer & Grest, <i>Dynamics of entangled linear polymer melts: A molecular-dynamics simulation</i>	The foundational bead-spring / Kremer-Grest polymer melt model; essential for entanglement, Rouse-to-reptation crossover, and generic polymer MD.	OUCI: ~3,572 citations.
1990	Mayo, Olafson & Goddard, <i>DREIDING: A generic force field for molecular simulations</i>	General-purpose force field often encountered in polymer, organic, and materials simulations. Not polymer-specific, but important background for force-field choices.	ACS: cited by ~5,948 publications.
1997	Paul, Smith & Yoon, <i>Static and Dynamic Properties of a n-C100H202 Melt from Molecular Dynamics Simulations</i>	Representative atomistic/united-atom polyethylene melt simulation; good example of comparing MD chain statistics and dynamics with experiment.	Classic atomistic PE melt paper.
1998	Sun, <i>COMPASS: An Ab Initio Force-Field Optimized for Condensed-Phase Applications</i>	Widely used all-atom force field for condensed-phase simulations of organic molecules and polymers; common in Materials Studio/Forcite workflows.	CoLab: ~5,474 citations.

Year	Paper	Why it matters	Citation signal
1998	Bennemann, Paul, Binder & Dunweg, <i>Molecular-dynamics simulations of the thermal glass transition in polymer melts</i>	Classic dense polymer melt glass-transition MD study; useful for alpha relaxation, cooling protocols, and mode-coupling-theory comparisons.	Classic glass-transition simulation paper.
2000	Borodin & Smith, <i>Molecular Dynamics Simulations of Poly(ethylene oxide)/LiI Melts. 2. Dynamic Properties</i>	Important polymer-electrolyte MD paper; useful for ion transport, segmental dynamics, and PEO-salt coordination.	ACS: cited by ~115 publications.
2001	Akkermans & Briels, <i>A structure-based coarse-grained model for polymer melts</i>	A key bottom-up coarse-graining paper; highlights structure reproduction versus thermodynamic consistency.	University of Twente/Scopus: ~125 citations.
2002	Muller-Plathe, <i>Coarse-graining in polymer simulation: From the atomistic to the mesoscopic scale and back</i>	Review-style entry point for systematic polymer coarse graining and backmapping.	Classic coarse-graining review.
2003	Reith, Putz & Muller-Plathe, <i>Deriving effective mesoscale potentials from atomistic simulations</i>	Canonical iterative Boltzmann inversion style reference for deriving coarse-grained potentials from atomistic data.	Widely used coarse-graining method paper.
2003	Smith et al., <i>A molecular dynamics simulation study of nanoparticle interactions in a model polymer-nanoparticle composite</i>	Early MD study of polymer-mediated nanoparticle interactions in a polymer matrix; useful for polymer nanocomposites.	ScienceDirect: cited by ~262 publications.

Year	Paper	Why it matters	Citation signal
2003	Starr, Douglas & Glotzer, <i>Origin of particle clustering in a simulated polymer nanocomposite and its impact on rheology</i>	Classic simulation paper on nanoparticle clustering, dispersion, shear, and rheology in polymer nanocomposites.	NIST-hosted publication; widely cited in PNC literature.
2004	Everaers et al., <i>Rheology and Microscopic Topology of Entangled Polymeric Liquids</i>	Primitive-path analysis foundation linking microscopic entanglement topology to rheology and plateau modulus.	Seminal Science paper.
2004	Faller, <i>Automatic coarse graining of polymers</i>	Clear review of automatic mapping and potential-optimization schemes for polymer coarse graining.	ScienceDirect: cited by ~102 publications.
2004	Paul & Smith, <i>Structure and dynamics of amorphous polymers: computer simulations compared to experiment and theory</i>	Broad review connecting atomistic and coarse-grained simulations with amorphous polymer experiments.	Broad review paper.
2006	Harmandaris et al., <i>Hierarchical Modeling of Polystyrene: From Atomistic to Coarse-Grained Simulations</i>	Practical multiscale workflow for polystyrene: atomistic model, coarse-graining, time mapping, and backmapping.	ACS: cited by ~334 publications.
2014	Liu et al., <i>Molecular dynamics simulation of dispersion and aggregation kinetics of nanorods in polymer nanocomposites</i>	Useful example for rod-like fillers, dispersion/aggregation kinetics, and CG polymer nanocomposite analysis.	ScienceDirect: cited by ~45 publications.
2017	Vogiatzis & Theodorou, <i>Multiscale Molecular Simulations of Polymer-Matrix Nanocomposites</i>	Open-access review of polymer-matrix nanocomposite simulations across length scales.	Springer: ~69 citations.

Year	Paper	Why it matters	Citation signal
2023	Shi, Qian & Lu, <i>Coarse-grained molecular dynamics simulation of polymers: Structures and dynamics</i>	Recent high-level review for CG polymer MD, including structure, dynamics, and chemically specific coarse-grained models.	Recent WIREs review.

2.4 Suggested Reading Path

1. Start with the generic melt baseline: Kremer & Grest 1990.
2. Add atomistic polymer examples: Theodorou & Suter 1985; Paul, Smith & Yoon 1997.
3. Learn force-field context: DREIDING 1990; COMPASS 1998.
4. Study glassy dynamics: Bennemann et al. 1998; Paul & Smith 2004.
5. Learn coarse graining: Akkermans & Briels 2001; Muller-Plathe 2002; Reith et al. 2003; Faller 2004; Harmandaris et al. 2006; Shi et al. 2023.
6. Study entanglement topology and rheology: Everaers et al. 2004.
7. Move to applications: Borodin & Smith 2000 for polymer electrolytes; Smith et al. 2003 and Starr et al. 2003 for polymer nanocomposites.

2.5 Tags to Use Later

- polymer-melt
- kremer-grest
- atomistic-md
- united-atom
- force-field
- coarse-graining
- backmapping
- glass-transition
- entanglement
- rheology
- polymer-electrolyte
- polymer-nanocomposite

2.6 To Expand

- Add BibTeX entries in papers/bib/.
- Add one Markdown note per paper in papers/notes/.
- Add a separate list for polymer crystallization, cross-linked polymers/epoxy, polymer membranes, and LAMMPS/RadonPy-specific workflows.

Chapter 3

Tools and References

This guide separates force-field models, simulation engines, polymer-building workflows, and research papers. A package may automate a workflow, but it does not make the underlying model valid for every polymer: check each force field’s chemical coverage and compare calculated properties with experiment or a trusted reference.

3.1 Classical Force Fields and Model Representations

Name	What it describes	Notes for polymer work
AMBER	A family of biomolecular force fields and simulation programs. The related GAFF extends AMBER-style parameters to common organic molecules.	GAFF: Wang, Wolf, Caldwell, Kollman & Case, “Development and testing of a general amber force field,” <i>Journal of Computational Chemistry</i> 25 (9), 1157–1174 (2004), DOI . Do not treat protein/nucleic-acid parameter sets as general polymer parameters; see AMBER documentation .

Name	What it describes	Notes for polymer work
CHARMM	A force-field family and simulation program, with parameter sets for biomolecules, lipids, and selected small molecules.	CHARMM can mean either the force-field family or the simulation engine. CGenFF: Vanommeslaeghe, Hatcher, Acharya et al., "CHARMM general force field: A force field for drug-like molecules compatible with the CHARMM all-atom additive biological force fields," <i>Journal of Computational Chemistry</i> 31 (4), 671–690 (2010), DOI. Check its chemical coverage before polymer use.
OPLS-AA	An all-atom potential developed for conformational energetics and organic-liquid properties.	Jorgensen, Maxwell & Tirado-Rives, "Development and testing of the OPLS all-atom force field on conformational energetics and properties of organic liquids," <i>Journal of the American Chemical Society</i> 118 (45), 11225–11236 (1996), DOI. Validate torsions, charges, and condensed-phase properties for the polymer of interest.
PCFF	A class-II, all-atom force-field family developed for synthetic polymers and related materials. The linked study applies PCFF to poly(epsilon-caprolactone); the CFF93 polycarbonate paper is a related foundational source.	Confirm that the installed parameter set covers the chemistry and properties being studied. PCFF and COMPASS are related CFF-family models, but their parameters are not interchangeable.

Name	What it describes	Notes for polymer work
GROMOS	A biomolecular force-field family and simulation package; its parameter sets are tuned for particular molecular classes and condensed-phase data.	Representative force-field reference: Oostenbrink, Villa, Mark & van Gunsteren, "A biomolecular force field based on the free enthalpy of hydration and solvation: the GROMOS force-field parameter sets 53A5 and 53A6," <i>Journal of Computational Chemistry</i> 25 (13), 1656–1676 (2004), DOI . Not a universal synthetic-polymer force field.
COMPASS	An all-atom, class-II force field parameterized for condensed-phase properties of organic, inorganic, and polymeric materials.	Sun, "COMPASS: An ab initio force-field optimized for condensed-phase applications—overview with details on alkane and benzene compounds," <i>The Journal of Physical Chemistry B</i> 102 (38), 7338–7364 (1998), DOI . Check installed version and parameter coverage.
ReaxFF	A bond-order reactive force field that can represent bond formation and breaking.	Foundational paper: van Duin, Dasgupta, Lorant & Goddard, "ReaxFF: A Reactive Force Field for Hydrocarbons," <i>The Journal of Physical Chemistry A</i> 105 (41), 9396–9409 (2001), DOI . Validate the parameter set for the relevant elements, bonding, and reaction conditions.
United-atom (UA)	A representation that combines many carbon-bound hydrogens with the carbon into one interaction site; TraPPE-UA is one named parameter family.	UA is a level of representation, not a force field by itself. Polar hydrogens and heteroatoms may remain explicit, depending on the model.

Name	What it describes	Notes for polymer work
Coarse-grained (CG)	A representation that groups atoms into larger interaction sites. Kremer & Grest, “Dynamics of entangled linear polymer melts: A molecular-dynamics simulation,” <i>The Journal of Chemical Physics</i> 92 , 5057–5086 (1990), introduced a generic bead-spring polymer-melt model; MARTINI 3 is a chemistry-informed CG force field.	CG models trade chemical detail for larger length and time scales. State the mapping and target properties; a generic model is not automatically transferable to a specific polymer.

3.2 Molecular-Dynamics and Modeling Software

Software	Main use	Links
LAMMPS	Open-source, parallel classical MD for polymers, atomistic materials, and coarse-grained systems. Supports many force-field styles, including class-II and reactive models.	Thompson et al., “LAMMPS—A flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales,” <i>Computer Physics Communications</i> 271 , 108171 (2022), DOI · documentation
GROMACS	Open-source, high-performance MD and trajectory analysis; commonly used for biomolecules and also used for polymer systems.	Abraham et al., “GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers,” <i>SoftwareX</i> 1–2 , 19–25 (2015), DOI · documentation · tutorials
NAMD	Parallel MD software designed for large biomolecular systems.	Phillips et al., “Scalable molecular dynamics on CPU and GPU architectures with NAMD,” <i>The Journal of Chemical Physics</i> 153 , 044130 (2020), DOI · user guide

Software	Main use	Links
CHARMM	Molecular simulation and analysis program that implements CHARMM force fields and other modeling methods.	Current documentation
BIOVIA Materials Studio	Commercial materials-modeling environment for building structures and running atomistic, mesoscale, and property calculations.	Product modules and capabilities depend on the licensed installation.

3.3 Polymer Construction, Simulation, and Analysis Workflows

Tool	What it helps with	Source
RadonPy	Open-source Python workflow that automates all-atom classical MD property calculations, including model construction, force-field assignment, equilibration checks, and post-processing. Check the repository for the current release and supported dependencies.	Hayashi, Shiomi, Morikawa & Yoshida, “RadonPy: automated physical property calculation using all-atom classical molecular dynamics simulations for polymer informatics,” <i>npj Computational Materials</i> 8 , 222 (2022), DOI · Documentation
pysimm	Open-source Python toolkit for building/modifying molecular systems, assigning force fields, and connecting workflows to LAMMPS and other engines; includes a polymer random-walk builder.	Fortunato & Colina, “pysimm: A Python Package for Simulation of Molecular Systems,” <i>SoftwareX</i> 6 , 7–12 (2017), DOI · current code/version: project documentation · v1.1.1 archive (2025)
PySoftK	Open-source Python package for building polymer chains and analyzing soft-matter simulations, including interfaces, aggregation/self-assembly, and intermolecular interactions. The cited v1.0 paper is primarily an analysis toolkit paper.	López-Ríos de Castro, Santana-Bonilla, Ziólek & Lorenz, “Automated Analysis of Soft Matter Interfaces, Interactions, and Self-Assembly with PySoftK,” <i>Journal of Chemical Information and Modeling</i> 65 (4), 1679–1684 (2025), DOI · Documentation

Tool	What it helps with	Source
SMiPoly	Rule-based generation of virtual polymer structures from monomers and polymerization reactions; the published rules target linear polymers, not arbitrary branched architectures.	Ohno, Hayashi, Zhang, Kaneko & Yoshida, "SMiPoly: Generation of a Synthesizable Polymer Virtual Library Using Rule-Based Polymerization Reactions," <i>Journal of Chemical Information and Modeling</i> 63 (17), 5539–5548 (2023), DOI · Documentation
SPACIER	Open-source design workflow integrating RadonPy all-atom property calculations and Bayesian optimization; demonstrated for optical polymer design.	Nanjo, Arifin, Maeda, Hayashi, Hatakeyama-Sato, Himeno, Hayakawa & Yoshida, "SPACIER: on-demand polymer design with fully automated all-atom classical molecular dynamics integrated into machine learning pipelines," <i>npj Computational Materials</i> 11 , 16 (2025), DOI · code · arXiv version
PEMD	Open-source Python framework unifying polymer construction, force-field parameterization, multiscale simulation, and electrolyte property analysis. The paper reports 656 successfully constructed homopolymers and screens 200 electrolyte systems.	Tan, Liang, Lu, Ji, Ji, Li & Hou, "PEMD: a high-throughput simulation and analysis framework for solid polymer electrolytes," <i>Digital Discovery</i> 5 (1), 193–202 (2026), DOI · code · archived release
MolPy	Open-source toolkit for inspectable, modular molecular-topology construction and editing, including reactive/cross-linking polymer workflows and structured interfaces for agent-assisted coding.	Li, Schwarz, Guo, Sun & Brandell, "MolPy: A Large Language Model-Friendly Toolkit for Reactive Topology Editing in Polymer Simulations," <i>Journal of Chemical Information and Modeling</i> 66 (13), 7562–7572 (2026), DOI

Tool	What it helps with	Source
PolyRapid	Adaptive automated annealing and screening demonstrated on 103 homopolymers; reports 95% equilibration within four 30 ns cycles and a public dataset for downstream modeling.	Smith, Ericson, Fantauzzo, Yong, Carbone & Troisi, “PolyRapid: automated high-throughput screening of polymers using a computational workflow,” <i>Soft Matter</i> 22 (33), 5480–5492 (2026), DOI · open full text
ADEPT / PolyGraphMT	ADEPT builds atomistic polymer models and computes properties; PolyGraphMT learns across property tasks and data fidelities using simulation, experiment, and group-contribution data.	Alosious, Liu, Xu, Liu, Zhang, Jiang & Luo, “ADEPT-PolyGraphMT: automated molecular simulation and multi-task multi-fidelity machine learning for polymer property generation and prediction,” <i>Digital Discovery</i> 5 (7), 3021–3043 (2026), DOI · preprint · code
PolyGraphPy	Open-source Python framework using DFTB+ data generation, Bayesian graph neural networks with uncertainty, and SELFIES/BRICS generative models for polymer property prediction and design.	Duarte, Venkatram, Cencer, Dumitrică & dos Santos, “PolyGraphPy: A unified Python framework for atomistic simulation and machine learning-driven polymer design,” arXiv:2606.06415v2 (2026; preprint as of 7 Oct 2026), arXiv · related journal article scheduled for 25 Oct 2026: <i>Computational Materials Science</i> 275 , 114985 · code

3.4 Force-Field Parameterization

Tool	Main use	Source
Sobtop	Desktop utility for supported small molecules that prepares force-field parameters and GROMACS topology files; the project page should be consulted for supported input formats and current release.	Official project/download page · cite software with the version/date actually used; no journal article located.

Tool	Main use	Source
LigParGen	Web server that assigns OPLS-AA parameters to supported organic ligands from a molecular structure/SMILES. It does not automatically validate polymer transferability.	Dodda, Cabeza de Vaca, Tirado-Rives & Jorgensen, “LigParGen web server: an automatic OPLS-AA parameter generator for organic ligands,” <i>Nucleic Acids Research</i> 45 (W1), W331–W336 (2017), DOI · Yale service
PolyParGen	Open-source workflow that partitions a polymer into supported fragments, obtains their parameters, and combines them into a polymer GROMACS topology; tested on P3EHT and F-P3EHT in chloroform.	Yabe, Mori, Ueda & Takeda, “Development of PolyParGen Software to Facilitate the Determination of Molecular Dynamics Simulation Parameters for Polymers,” <i>Journal of Computer Chemistry, Japan—International Edition</i> 5 , 2018-0034 (2019), DOI · article

3.5 Machine-Learned Force Fields (MLFFs) and Polymer-Property Papers

Paper	What it contributes	Publication status
SimPoly: Simulation of Polymers with Machine Learning Force Fields Derived from First Principles	Introduces Vivace, an ML force field trained from first-principles data, and PolyArena, an experimental bulk-property benchmark spanning 130 polymers; reports density and glass-transition predictions.	Simm, Hélie, Schulz, Chen, Simeon, Kuzina, Martinez-Baez, Gasparotto, Tocci, Chen, Li, Cheng, Wang, Nguyen, Smith & Sun, arXiv:2510.13696 (v1 submitted 15 Oct 2025; preprint), arXiv DOI · code and model
On-the-Fly Machine-Learned Force Fields for High-Fidelity Polymer Glass Transition Simulations	Adaptively calls first-principles calculations when configurations leave the model’s confidence domain; demonstrates near-quantum Tg simulations on 12 polymers and reports roughly six orders lower cost than AIMD.	Srivastava, Agarwal, Shukla, Sahu & Ramprasad, <i>Journal of Chemical Information and Modeling</i> 66 (17), 11129–11139 (2026), DOI · arXiv:2601.17137

Paper	What it contributes	Publication status
Active Learning of a Neural Network Potential for Large-Scale Atomistic Simulations of Polymer Electrolyte Membranes	Active-learning deep potential for Nafion over hydration states; enables 10,000–20,000-atom simulations to 31 ns and tests transfer to hydration level $\lambda = 24$ beyond the training range.	Yoshimoto, Matsumura, Yamazaki, Iwasaki, Nakao & Sakai, <i>The Journal of Physical Chemistry B</i> 130 (36), 9326–9337 (2026), DOI
How Long Is Long Enough? Extrapolation of Machine-Learning Interatomic Potentials for Oligomeric and Polymeric Systems	Controlled n-alkane training study (n=1–8): transfer improves when local chemical environments converge; careful neighbor-list construction improves intermolecular-energy learning.	Hooven, Lin, Carroll & Cersonsky, <i>Journal of Chemical Theory and Computation</i> 22 (13), 6917–6926 (2026), DOI
From Oligomers to Entangled Polymers: How to Train a Transferable Machine Learning Interatomic Potential	Compares descriptors and active learning for polyethylene; reports ACE transfer from small oligomers and evaluates non-bonded interactions against OPLS-AA trajectories.	Fischer & Heuer, arXiv:2608.01162 (submitted 2 Aug 2026; preprint), arXiv
Glass Transition Prediction of Binary Copolymers Across Large Chemical Spaces Using Machine Learning and Physics-Based Modeling	Composition-aware models trained on experimental data report RMSE about 14 K and R-squared about 0.98; screens about 148,000 binary copolymer compositions and validates with MD.	Bhati, Afzal, Chew, Browning & Halls, <i>Polymers</i> 18 (14), 1727 (2026), DOI
Rapid, Accurate and Reproducible Prediction of the Glass Transition Temperature Using Ensemble-Based Molecular Dynamics Simulation	Quantifies replica uncertainty for six cross-linked epoxy resins; concurrent-temperature simulation agrees with sequential cooling, and 10 or more replicas support 95% confidence intervals below 20 K.	Suter, Müller, Vassaux, Anastasiou, Simmons, Tilbrook & Coveney, <i>Journal of Chemical Theory and Computation</i> 21 (3), 1405–1421 (2025), DOI
PolyRapid: Automated High-Throughput Screening of Polymers Using a Computational Workflow	Screens 103 homopolymers with adaptive annealing; reports 95% equilibrated within four 30 ns cycles, 100% within nine, and ML classification for density/T _g .	Smith, Ericson, Fantauzzo, Yong, Carbone & Troisi, <i>Soft Matter</i> 22 (33), 5480–5492 (2026), DOI

Paper	What it contributes	Publication status
Dissipative Particle Dynamics Modeling in Polymer Science and Engineering	Review of DPD methods, mapping and polymer applications spanning blends, solutions, self-assembly, interfaces, membranes, rheology, and composites.	Nikkhah & Vandichel, <i>WIREs Computational Molecular Science</i> 15 (2), e70018 (2025), DOI
Atomistic Polymer Modeling: Recent Advances and Challenges in Building and Parametrization Workflows	Reviews synthetic-polymer system-building/parameterization gaps, FAIR practice, cheminformatics, direct chemical perception, and neural-network workflows.	Turney & Matta, <i>Macromolecules</i> 58 (21), 11509–11522 (2025), DOI
ADEPT-PolyGraphMT: Automated Molecular Simulation and Multi-Task Multi-Fidelity Machine Learning for Polymer Property Generation and Prediction	Integrates atomistic simulation with curated experimental and group-contribution values into ~62,000 records for 28 properties; benchmarks multi-task and multi-fidelity learning.	Alosious, Liu, Xu, Liu, Zhang, Jiang & Luo, <i>Digital Discovery</i> 5 (2026), DOI · preprint

3.6 Polymer Discovery Agents and Representations

Resource	Scope	Source / status
Polymer-Agent	LLM-driven terminal workflow for polymer property prediction, property-guided structure generation, and candidate modification; a research preprint, not a validated general-purpose autonomous lab.	Nigam, Chandrasekhar & Farimani, “PolyAgent: Large Language Model Agent for Polymer Design,” arXiv:2601.16376 (2026; preprint), arXiv · repository
PolyJarvis	Natural-language agent that connects an LLM to RadonPy through MCP to build, parameterize, equilibrate, and analyze all-atom polymer simulations. The authors report mixed accuracy across four polymers and eight comparisons.	Zhao, Chandrasekhar & Farimani, “PolyJarvis: LLM Agent for Autonomous Polymer MD Simulations,” arXiv:2604.02537 (2026; preprint), arXiv · repository

Resource	Scope	Source / status
ToPolyAgent	Natural-language multi-agent workflow for coarse-grained bead-spring LAMMPS simulations, including linear, ring, brush, star, and dendritic topologies.	Ding, Carrillo & Do, “ToPolyAgent: AI agents for coarse-grained bead-spring topological polymer simulations,” <i>Digital Discovery</i> 5 (2), 901–909 (2026), DOI · repository
MDCrow	General LLM-assisted MD workflow with 40 tools for simulation setup, file handling, analysis, and literature retrieval. The published benchmark focuses on protein simulations, not synthetic-polymer workflows.	Campbell, Cox, Medina, Watterson & White, “MDCrow: automating molecular dynamics workflows with large language models,” <i>Machine Learning: Science and Technology</i> 7 (2), 025037 (2026), DOI · 2025 preprint · code
G-BigSMILES 2.0	Stochastic, generative polymer representation that encodes distributions and complex architectures; can generate molecular ensembles and graph-learning inputs. This is a representation, not a simulation engine.	Tian, Zaldivar & Sun, “G-BigSMILES 2.0: A Generative Representation for Complex Polymer Architectures and Machine Learning,” APS Global Physics Summit 2026, oral presentation (16 Mar 2026), abstract

3.7 Notes on Scope

- A **force field** defines interactions; an **MD engine** integrates equations of motion; a **workflow package** automates some steps around an engine. These are related, but not interchangeable.
- UA and CG describe levels of molecular representation. TraPPE-UA, Kremer–Grest, and MARTINI are examples of specific parameterized models within those representations.
- When reporting software-based results, record the exact software release/commit, force-field family and parameter-set version, input structure, mapping, and validation protocol. A package or force-field family name alone is not enough to reproduce a calculation.
- For a new polymer chemistry, verify atom typing, charges, bonded parameters, non-bonded interactions, and any mapping choices. Report the validation data and simulation protocol with the results.
- Preprints, conference presentations, and peer-reviewed articles are labeled separately above. Recheck publication status when citing fast-moving 2026 work.